

Implementation Roadmap

ServiceGuard AI — From Architecture to MVP

Phase 0 — Product and dataset preparation

- Finalize domain vocabulary.
- Generate/validate telemetry dataset.
- Create runbook/SOP assets.
- Finalize ERD and API contracts.

Phase 1 — Java/PostgreSQL foundation

- Spring Boot project.
- PostgreSQL migrations.
- Customer/Application/Service/Instance entities.
- Authentication and RBAC.

Phase 2 — Telemetry and incidents

- Telemetry ingestion/query APIs.
- SLA configuration.
- Deterministic breach detection.
- Incident lifecycle.

Phase 3 — Python intelligence service

- FastAPI service.
- Internal authentication.
- Telemetry retrieval client.
- ML inference contract.

Phase 4 — ML

- EDA.
- Feature engineering.
- Baseline model.
- Evaluation.
- Model persistence/versioning.
- Inference endpoint.

Phase 5 — RAG

- Runbook ingestion.
- Chunking.
- Embeddings.
- pgvector.

- Metadata filtering.
- Evidence return.

Phase 6 — Agent

- LangGraph/state.
- Telemetry tool.
- Dependency tool.
- RAG tool.
- ML tool.
- Structured recommendation.
- Guardrails.

Phase 7 — Product UI

- Dashboard.
- Service detail.
- Incident center.
- AI copilot.
- Evidence panel.
- Approval workflow.
- Audit view.

Phase 8 — Production hardening

- Redis rate limiting/cache.
- Kafka if required.
- Docker.
- CI/CD.
- Observability.
- Load testing.
- Security review.

Definition of Done

- One complete degradation scenario works end-to-end.
- Recommendation is evidence-backed.
- Human approval is enforced.
- AI trace is auditable.
- Automated tests pass.
- System runs locally through Docker.
- Documentation and README are complete.

Golden demo scenario

```
Service degrades
↓
Incident detected
```

↓
Risk = HIGH
↓
Runbook retrieved
↓
AI explains cause/evidence
↓
Recommendation
↓
Engineer approves
↓
Audit record